Adilkhan Salkimbayev

Applied Mathematics & Scientific Machine Learning Researcher
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EDUCATION

Kazakh-British Technical University (KBTU)
B.Sc. in Automation and Control | GPA: 3.58 | Aug 2023 – July 2027 (Expected)

- **Relevant Coursework:** Linear Algebra for Engineers, Differential Equations, Calculus of Complex Variables
- **Military Department:** Sergeant of the Reserve (Completed June 2025).

PUBLICATIONS AND PREPRINTS

- **A. Salkimbayev.** "Structural Guarantees in the Approximation of Model Predictive Control Laws: A Shape-Constrained Polynomial Read-Out." *Resubmitted, Journal of Process Control.*
- **A. Salkimbayev, K.S. Haider.** "Robust Adaptive Kolmogorov–Arnold Neural Control." *Major Revision, International Journal of Adaptive Control and Signal Processing.*
- **A. Salkimbayev.** "Leray–Hopf Weak Solutions of the 2D Navier–Stokes Equations on the Torus: An Expository Construction." *Expository article; preprint, 34 pp.*
- **Further preprints.** Three programs, twenty papers in all, September–October 2026: "The Euler–Arnold Program"; "Transport and the Supercriticality Gap in 3D Navier–Stokes"; "Strained Vortex Tubes and the Covering of Collapse in 3D Navier–Stokes." *Produced with substantial AI assistance under my direction; not yet independently reviewed. Listed with links at hecatetdm.tech/cv.*

RESEARCH EXPERIENCE

Independent Undergraduate Researcher | Kazakh-British Technical University

Functional-Analytic Control Architecture & PDEs (Current)

- Developing a novel functional-analytic control architecture drawing upon operator theory and methodologies from Brezis, Evans and Stein.
- Translating control frameworks into rigorous mathematical formulations applicable to the study of partial differential equations and fluid dynamics.
- Wrote a 34-page expository construction of Leray–Hopf weak solutions to the 2D Navier–Stokes equations on the torus, proving uniqueness in two dimensions and showing where its three-dimensional strong-solution estimate stops.

Adaptive Control & Nonlinear Dynamical Systems (Jan 2026 – Jul 2026)

- Developed a learning-based adaptive control architecture combining Recursive Least Squares with symbolic Kolmogorov–Arnold Networks (KANs).
- Derived rigorous Lyapunov stability guarantees (ISS / UUB) over the physically realizable operating envelope, with a certificate that requires no persistent excitation.
- Evaluated the algorithm on a quadruple-tank benchmark via a 50-seed Monte-Carlo campaign in Python, demonstrating a branch-free per-step runtime bounded a priori and a 7% aggregate tracking improvement over LTV-MPC.

Constant-Time Explicit Control Laws from MPC (Nov 2025 – Apr 2026)

- Designed a certifiable explicit replacement for Linear Time-Varying MPC on the quadruple-tank benchmark in its non-minimum-phase configuration: an analytic feedforward term, a detuned

LQR core, and a learned correction behind a gate that vanishes quadratically at the setpoint, making zero steady-state offset and local exponential stability structural rather than dependent on fit quality.
- Built a two-step symbolic read-out in which a Kolmogorov–Arnold Network (KAN) supplies only the monomial support and the coefficients are re-estimated by a convex quadratic program under affine negative-feedback constraints — cutting a sign pathology present on up to 11% of the operating box to at most 0.6%, for 0.3–1.7 percentage points of accuracy.
- Established the constant-time claim structurally in software-in-the-loop — a fixed chain of 38–211 multiply-accumulate operations with no data-dependent branching, stable under ±20% parameter perturbation in 96.5–100% of Monte-Carlo trials — and reported the negative result that the KAN-selected support is no better than orthogonal matching pursuit at a matched arithmetic budget, and worse in the non-minimum-phase regime.

ADVANCED INDEPENDENT STUDY

Self-directed graduate sequence in analysis and PDE, directed toward the rigorous study of fluid dynamics:

- **Completed:** *Introduction to Real Analysis* (William F. Trench); *Functional Analysis, Sobolev Spaces and Partial Differential Equations* (Haim Brezis), Chs. 1–9 — including semigroup theory (Hille–Yosida) and the variational formulation of elliptic boundary-value problems; *Partial Differential Equations* (Lawrence C. Evans), Chs. 5–6 — Sobolev spaces and second-order elliptic theory.
- **In progress:** *Real Analysis: Measure Theory, Integration, and Hilbert Spaces* (Elias M. Stein & Rami Shakarchi), Chs. 1–3.
- **Planned (Fall 2026):** *Measure Theory and Fine Properties of Functions* (Lawrence C. Evans & Ronald F. Gariepy); *Singular Integrals and Differentiability Properties of Functions* (Elias M. Stein).

TECHNICAL & COMPUTATIONAL SKILLS

Mathematics & Control Systems:
Operator theory, partial differential equations, semigroup theory, Lyapunov stability analysis, adaptive control, nonlinear control systems

Scientific Machine Learning:
Kolmogorov–Arnold Networks, system identification, embedded SciML.

Programming & Hardware: Python (NumPy, SciPy, matplotlib, pandas, PyTorch), C/C++, STM32 CubeIDE, LaTeX (VSCode).